MAT 3007 – Optimization

Optimality Condition for Constrained Problems

Lecture 18

Instructor: Junchi Yang

CUHK-Shenzhen

Multivariable Calculus and Linear Algebra

Eigenvalues and Eigenvectors

Definition: Eigenvalues and Eigenvectors

Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be a square matrix. A scalar $\lambda \in \mathbb{R}$ and a non-zero vector $\boldsymbol{x} \in \mathbb{R}^n$ are called an eigenvalue and an eigenvector of $\boldsymbol{A}$, respectively, if

$$\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x}.$$

- The eigenvector $\boldsymbol{x}$ indicates a direction that remains unchanged under the transformation $\boldsymbol{A}$.
- The eigenvalue λ represents the scaling factor for the eigenvector $\boldsymbol{x}$.

Eigen-Decomposition

Definition: Eigen-Decomposition

If $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is a diagonalizable matrix, it can be expressed as:

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^{-1},$$

where:

- $\boldsymbol{Q}$ is a matrix whose columns are the eigenvectors of $\boldsymbol{A}$.
- $\boldsymbol{\Lambda}$ is a diagonal matrix whose entries are the eigenvalues of $\boldsymbol{A}$.

- Any symmetric matrix $\boldsymbol{A}$ is diagonalizable. Moreover, its eigenvalues are real and eigenvectors corresponding to distinct eigenvalues are orthogonal. $Q^T Q = Q Q^T = I$

Finding Eigenvalues and Eigenvectors

Characteristic Equation

The eigenvalues of $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ are found by solving the characteristic equation:

$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0,$$

where $\boldsymbol{I}$ is the identity matrix of size $n \times n$.

Finding Eigenvalues and Eigenvectors

Characteristic Equation

The eigenvalues of $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ are found by solving the characteristic equation:

$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0,$$

where $\boldsymbol{I}$ is the identity matrix of size $n \times n$.

Steps to Find Eigenvectors

Once eigenvalues λ are determined:

1. Solve $(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}$ to find $\boldsymbol{x}$.
2. Ensure $\boldsymbol{x} \neq \boldsymbol{0}$ (non-trivial solutions).

Trace and Determinant of a 2×2 Matrix

Statement: For a 2×2 matrix $\boldsymbol{A}$, the trace and determinant are related to its eigenvalues λ_1 and λ_2 as follows:

$$\mathrm{tr}(\boldsymbol{A}) = \lambda_1 + \lambda_2, \quad \det(\boldsymbol{A}) = \lambda_1 \lambda_2.$$

Trace and Determinant of a 2×2 Matrix

Statement: For a 2×2 matrix $\boldsymbol{A}$, the trace and determinant are related to its eigenvalues λ_1 and λ_2 as follows:

$$\mathrm{tr}(\boldsymbol{A}) = \lambda_1 + \lambda_2, \quad \det(\boldsymbol{A}) = \lambda_1 \lambda_2.$$

Why?

- The characteristic polynomial of $\boldsymbol{A}$ is:

$$p(\lambda) = \det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \lambda^2 - \mathrm{tr}(\boldsymbol{A})\lambda + \det(\boldsymbol{A}).$$

- Eigenvalues λ_1 and λ_2 are roots of $p(\lambda)$:

$$p(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1 \lambda_2.$$

- Comparing coefficients:

$$\mathrm{tr}(\boldsymbol{A}) = \lambda_1 + \lambda_2, \quad \det(\boldsymbol{A}) = \lambda_1 \lambda_2.$$

Bounding the Quadratic Form

Theorem

Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Then:

$$\lambda_{\min}(\boldsymbol{A})\|x\|^2 \leq x^\top \boldsymbol{A} x \leq \lambda_{\max}(\boldsymbol{A})\|x\|^2 \quad \forall x \in \mathbb{R}^n,$$

where $\lambda_{\min}(\boldsymbol{A})$ and $\lambda_{\max}(\boldsymbol{A})$ are the smallest and largest eigenvalues of A.

Proof: Consider eigen-decomp. of A. Because A is symmetric $A = Q \Lambda Q^T$ with $Q^TQ = QQ^T = I$. For any x,

$$x^TAx = x^TQ\Lambda \underbrace{Q^Tx}_{y} = y^T\Lambda y = \sum \lambda_i y_i^2 \leq \lambda_{max} \sum y_i^2 = \lambda_{max}\|y\|^2 = \lambda_{max}\|x\|^2$$

Note $\|y\|^2 = \|Q^Tx\|^2 = x^TQQ^Tx = \|x\|^2$

Recap

Indefiniteness and Saddle Points

Definition: Stationary Points and Saddle Points

- A point $\boldsymbol{x}$ satisfying $\nabla f(\boldsymbol{x}) = \mathbf{0}$ is called critical point or stationary point.
- A stationary point is called saddle point if it is neither a local minimizer nor a local maximizer.

Indefiniteness and Saddle Points

Definition: Stationary Points and Saddle Points

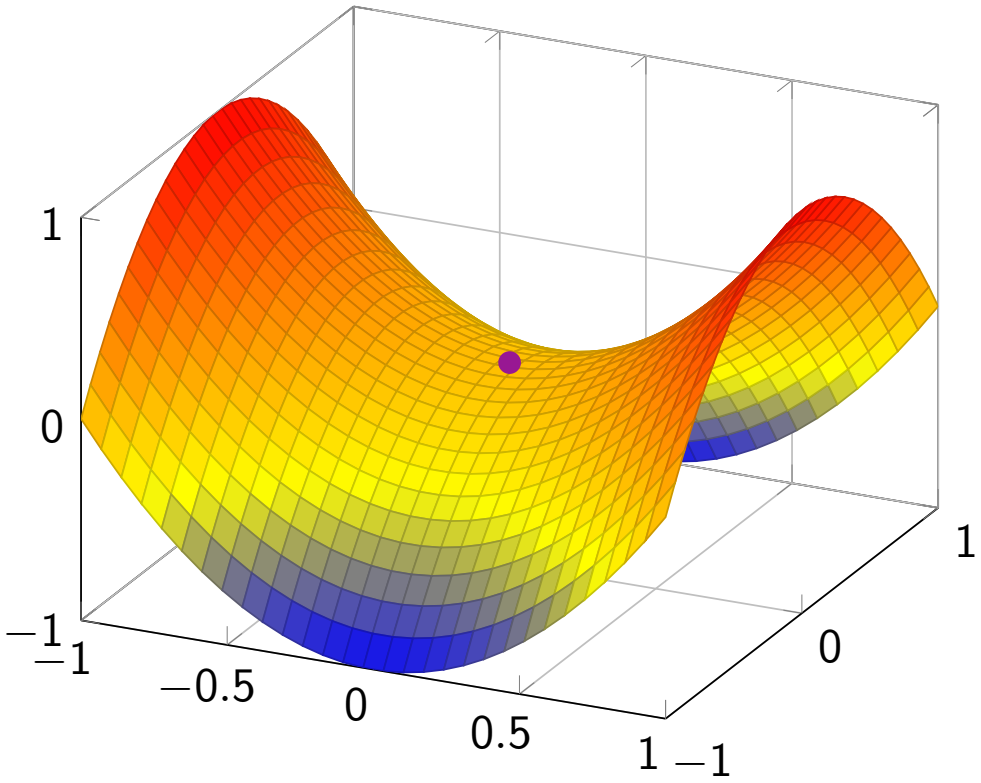
- A point $\boldsymbol{x}$ satisfying $\nabla f(\boldsymbol{x}) = \mathbf{0}$ is called critical point or stationary point.
- A stationary point is called saddle point if it is neither a local minimizer nor a local maximizer.

Corollary: A Sufficient Condition for Saddle Points

Suppose that $\boldsymbol{x}^*$ is a stationary point and that the Hessian $\nabla^2 f(\boldsymbol{x}^*)$ is indefinite, then $\boldsymbol{x}^*$ is a saddle point.

- It is a sufficient but not necessary condition. Consider $f(x) = x^3$ at $x = 0$.

Example: Saddle Point

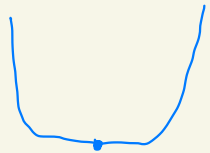


Plot of the function $f(\boldsymbol{x}) = x_1^2 - x_2^2$.

- The gradient is $\nabla f(\boldsymbol{x}) = (2x_1, -2x_2)^\top$ and $\boldsymbol{x}^* = (0,0)^\top$ is the only stationary point of f. Since $\nabla^2 f(\boldsymbol{x}^*)$ is indefinite, $\boldsymbol{x}^*$ has to be a saddle point.

Question 1: Any example that satifies SONC but not SOSC?

All non-strict local mimizer



Question 2: Is there any strict local min that does not satify SOSC?

Yes. $f(x,y) = x^2$ at point $(0,0)$

$\nabla^2 f(0,0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$. This is PSD but not PD.

However, $(0,0)$ is a strict local min.

Summary for Unconstrained Problems

- All stationary points (FONC) are candidates for local minimizers. (not sufficient).
- SONC can further remove some stationary points that are not local minimizers, including local maximizers and saddle points with indefinite Hessian.

Summary for Unconstrained Problems

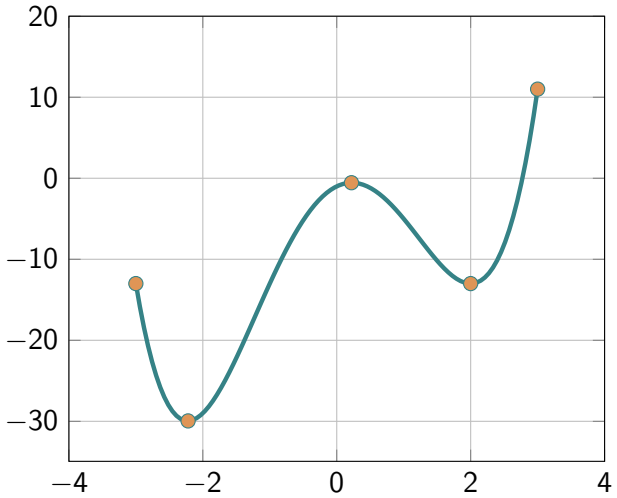
- All stationary points (FONC) are candidates for local minimizers. (not sufficient).
- SONC can further remove some stationary points that are not local minimizers, including local maximizers and saddle points with indefinite Hessian.
- SOSC can be used to sufficiently identify some strict local minimizers, not all.
- For non-strict local minimizers, strict local minimizers satisfying SONC, and saddle points with PSD Hessian (consider x^3 at $x = 0$), SONC is satisfied in these cases. So far we do not have ways to distinguish them.

Summary for Unconstrained Problems

- All stationary points (FONC) are candidates for local minimizers. (not sufficient).
- SONC can further remove some stationary points that are not local minimizers, including local maximizers and saddle points with indefinite Hessian.
- SOSC can be used to sufficiently identify some strict local minimizers, not all.
- For non-strict local minimizers, strict local minimizers satisfying SONC, and saddle points with PSD Hessian (consider x^3 at $x = 0$), SONC is satisfied in these cases. So far we do not have ways to distinguish them.

A special case: If a problem has only one stationary point and a finite optimal solution, then the stationary point must be the unique global solution.

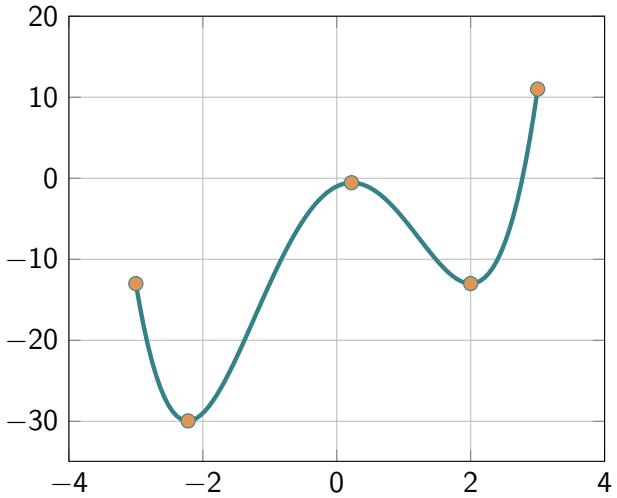
$$\min_{x\in\mathbb{R}} f(x) := x^4 - 9x^2 + 4x - 1 \quad \text{subject to} \quad x \in [-3, 3].$$



At the boundary ($x = 3$), the FONC is not satisfied:

$$f'(3) = 58 > 0.$$

$$\min_{x\in\mathbb{R}} f(x) := x^4 - 9x^2 + 4x - 1 \quad \text{subject to} \quad x \in [-3, 3].$$



At the boundary ($x = 3$), the FONC is not satisfied:

$$f'(3) = 58 > 0.$$

$\rightsquigarrow$ It means that the developed FONC ($\nabla f(\boldsymbol{x}) = \mathbf{0}$) for unconstrained problems is not enough for constrained problems.

Feasible Directions

Definition: Feasible Direction

Given $\boldsymbol{x} \in \Omega$, we call $\boldsymbol{d}$ a feasible direction at $\boldsymbol{x}$ if there exists $\bar{t} > 0$ such that $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all $0 \leq t \leq \bar{t}$.

Feasible Directions

Definition: Feasible Direction

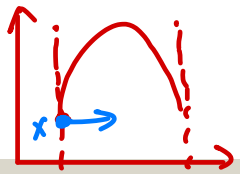
Given $\boldsymbol{x} \in \Omega$, we call $\boldsymbol{d}$ a feasible direction at $\boldsymbol{x}$ if there exists $\bar{t} > 0$ such that $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all $0 \leq t \leq \bar{t}$.

Example:

- If $\Omega = \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\}$, then all feasible directions at $\boldsymbol{x}$ are given by $\{\boldsymbol{d} : \boldsymbol{A}\boldsymbol{d} = \boldsymbol{0}\}$.
- If $\Omega = \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}\}$, then the feasible directions at $\boldsymbol{x}$ are given by $\{\boldsymbol{d} : \boldsymbol{a}_i^\top \boldsymbol{d} \leq 0 \text{ for all } i \text{ with } \boldsymbol{a}_i^\top \boldsymbol{x} = b_i\}$.

FONC for Constrained Problems

Example



at point x. feasible directions $\{d : d \geq 0\}$

$\nabla f(x) \geq 0 \quad \Rightarrow \quad \nabla f(x) \cdot d \geq 0$

Theorem: FONC for Constrained Problems

Let $\boldsymbol{x}^*$ be a local minimum of $\min_{\boldsymbol{x} \in \Omega} f(\boldsymbol{x})$. Then for any feasible direction $\boldsymbol{d}$ at $\boldsymbol{x}^*$, we must have $\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \geq 0$.

Proof:

- $\rightsquigarrow$ As before by Taylor expansion and using $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all t sufficiently small.

Remark:

- In the unconstrained case, all directions are feasible. Thus, we must have $\nabla f(\boldsymbol{x}^*) = \boldsymbol{0}$.

Theorem: FONC for Constrained Problems

Let $\boldsymbol{x}^*$ be a local minimum of $\min_{\boldsymbol{x}\in\Omega} f(\boldsymbol{x})$. Then for any feasible direction $\boldsymbol{d}$ at $\boldsymbol{x}^*$, we must have $\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \geq 0$.

Proof:

$\rightsquigarrow$ As before by Taylor expansion and using $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all t sufficiently small.

Remark:

- In the unconstrained case, all directions are feasible. Thus, we must have $\nabla f(\boldsymbol{x}^*) = \boldsymbol{0}$.

$\rightsquigarrow$ This is also true, when $\boldsymbol{x}^*$ lies in the interior of Ω.

proof: Assume $\nabla f(x^*)^T d < 0$ for some d. By Taylor's theorem,

$$f(x^* + td) = f(x^*) + t\nabla f(x^*)^T d + O(t\|d\|)$$

$\nabla f(x^*)^T d < 0$

When t is small enough, sum of last two terms is smaller than 0,

$$\rightarrow f(x^* + td) < f(x^*)$$

Also, since d is a feasible direction, $x^* + td \in \Omega$ as t is small enough. $\rightarrow$ contradiction

□

Remark: (a) unconstrained problem.

any d is a feasible direction

Then $d = e_i$ and $-e_i$ are feasible direction

$\Rightarrow \nabla f(x^*)^T e_i \geq 0$ and $-\nabla f(x^*)^T e_i \geq 0$

$\Leftrightarrow [\nabla f(x^*)]_i \geq 0$ and $[\nabla f(x^*)]_i \leq 0$

$\Leftrightarrow [\nabla f(x^*)]_i = 0 \quad \forall i \Leftrightarrow \nabla f(x^*) = 0$

(b) If x^* is an interior point of Ω, then

all directions are feasible direction (why?) $\rightarrow \nabla f(x^*) = 0$

An Alternative View: Descent Directions

Definition: Descent Direction

Let f be continuously differentiable. Then $\boldsymbol{d}$ is called a descent direction at $\boldsymbol{x}$ if and only if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.

Remark: By Taylor's theorem,

$$f(\boldsymbol{x} + t\boldsymbol{d}) = f(\boldsymbol{x}) + t\nabla f(\boldsymbol{x})^\top \boldsymbol{d} + o(t\|\boldsymbol{d}\|)$$

- If $\boldsymbol{d}$ is a descent direction at $\boldsymbol{x}$, there exists $\bar{t} > 0$ such that $f(\boldsymbol{x} + t\boldsymbol{d}) < f(\boldsymbol{x})$ for all $0 < t \leq \bar{t}$. That is, we can decrease the function value by moving in the direction of $\boldsymbol{d}$.

If we denote the set of feasible directions at $\boldsymbol{x}$ by $S_\Omega(\boldsymbol{x})$ and the set of descent directions at $\boldsymbol{x}$ by $S_D(\boldsymbol{x})$, then the first order necessary condition can be written as:

$$S_\Omega(\boldsymbol{x}^*) \cap S_D(\boldsymbol{x}^*) = \emptyset$$

$\rightsquigarrow$ There are no feasible descent directions.

General Optimality Conditions

We want to use the notion of feasible directions to obtain optimality conditions for the nonlinear programs:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad \forall\ i = 1,\dots,m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\ j = 1,\dots,p. \end{aligned}$$

We will also work with the following index sets:

Definition: Active and Inactive Set

At a point $\boldsymbol{x} \in \Omega$, the set $\mathcal{A}(\boldsymbol{x}) := \{i : g_i(\boldsymbol{x}) = 0\}$ denotes the set of active constraints. The set of inactive constraints is then given by $\mathcal{I}(\boldsymbol{x}) := \{i : g_i(\boldsymbol{x}) < 0\}$.

$\rightsquigarrow$ We first consider linear constraints as special case.

Optimality Conditions: Linear Constraints

Problems with Linear Inequality Constraints

We now first consider an inequality constrained problem:

$$A = \begin{bmatrix} - & a_1^\top & - \\ & \vdots & \\ - & a_n^\top & - \end{bmatrix}$$

$$\text{minimize}_{\boldsymbol{x}} \; f(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}. \tag{1}$$

How can we express the necessary optimality conditions?

Theorem: FONC for Linearly Constrained Problems

If $\boldsymbol{x}^*$ is a local minimum of (1), then there exists some $\boldsymbol{y} \geq \boldsymbol{0}$ with

$$\nabla f(\boldsymbol{x}^*) + \boldsymbol{A}^\top \boldsymbol{y} = \boldsymbol{0}$$
$$y_i \cdot (\boldsymbol{a}_i^\top \boldsymbol{x}^* - b_i) = 0 \quad \forall\, i,$$

$\nabla f(x^*)$ can be represented as linear combination of $a_1, \ldots, a_n$

where $\boldsymbol{a}_i^\top$ is the ith row of $\boldsymbol{A}$.

complementary slackness

Proof

By the FONC for constrained problems:

$$\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \geq 0, \quad \forall \boldsymbol{d} \in S_\Omega(\boldsymbol{x}^*).$$

This is equivalent to

$$\left[\min_{\boldsymbol{d}\in\mathbb{R}^n} \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d}, \quad \text{s.t.} \quad \boldsymbol{d} \in S_\Omega(\boldsymbol{x}^*)\right] \geq 0,$$

Proof

By the FONC for constrained problems:

$$\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \geq 0, \quad \forall \boldsymbol{d} \in S_\Omega(\boldsymbol{x}^*).$$

This is equivalent to

$$\left[\min_{\boldsymbol{d}\in\mathbb{R}^n} \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d}, \quad \text{s.t.} \quad \boldsymbol{d} \in S_\Omega(\boldsymbol{x}^*)\right] \geq 0,$$

which is further equivalent to

$$\left[\min_{\boldsymbol{d}\in\mathbb{R}^n} \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d}, \quad \text{s.t.} \quad \boldsymbol{a}_i^\top \boldsymbol{d} \leq 0, \forall i \in \mathcal{A}(\boldsymbol{x}^*)\right] \geq 0,$$

By the FONC for constrained problems:

$$\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \geq 0, \quad \forall \boldsymbol{d} \in S_\Omega(\boldsymbol{x}^*).$$

This is equivalent to

$$\left[\min_{\boldsymbol{d}\in\mathbb{R}^n} \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d}, \quad \text{s.t.} \quad \boldsymbol{d} \in S_\Omega(\boldsymbol{x}^*)\right] \geq 0,$$

which is further equivalent to

$$\left[\min_{\boldsymbol{d}\in\mathbb{R}^n} \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d}, \quad \text{s.t.} \quad \boldsymbol{a}_i^\top \boldsymbol{d} \leq 0, \forall i \in \mathcal{A}(\boldsymbol{x}^*)\right] \geq 0,$$

Define

$$\boldsymbol{C} = \left(\boldsymbol{a}_i^\top\right)_{i\in\mathcal{A}(\boldsymbol{x}^*)}$$

The above optimization problem is further equivalent to

$$\left[\min_{\boldsymbol{d}\in\mathbb{R}^n} \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d}, \quad \text{s.t.} \quad \boldsymbol{C}\boldsymbol{d} \leq \boldsymbol{0}\right] \geq 0,$$

Proof Continued

This is a linear programming. Taking its dual,

$$\max_{\boldsymbol{z}\in\mathbb{R}^{|\mathcal{A}(\boldsymbol{x}^*)|}} \quad 0, \quad \text{s.t.} \quad \boldsymbol{z} \leq \boldsymbol{0},\ \boldsymbol{C}^\top \boldsymbol{z} = \nabla f(\boldsymbol{x}^*).$$

Proof Continued

This is a linear programming. Taking its dual,

$$\max_{\boldsymbol{z}\in\mathbb{R}^{|\mathcal{A}(\boldsymbol{x}^*)|}} \quad 0, \quad \text{s.t.} \quad \boldsymbol{z} \leq \boldsymbol{0},\ \boldsymbol{C}^\top \boldsymbol{z} = \nabla f(\boldsymbol{x}^*).$$

Since the primal is lower bounded and feasible, then the dual problem is feasible. Therefore, there exists a feasible $\boldsymbol{z}$:

$$\boldsymbol{z} \leq \boldsymbol{0},\ \nabla f(\boldsymbol{x}^*) = \boldsymbol{C}^\top \boldsymbol{z} = \sum_{k=1}^{|\mathcal{A}|} \boldsymbol{a}_{\mathcal{A}_k} z_k.$$

Proof Continued

This is a linear programming. Taking its dual,

$$\max_{\boldsymbol{z}\in\mathbb{R}^{|\mathcal{A}(\boldsymbol{x}^*)|}} \quad 0, \quad \text{s.t.} \quad \boldsymbol{z} \leq \boldsymbol{0},\ \boldsymbol{C}^\top \boldsymbol{z} = \nabla f(\boldsymbol{x}^*).$$

Since the primal is lower bounded and feasible, then the dual problem is feasible. Therefore, there exists a feasible $\boldsymbol{z}$:

$$\boldsymbol{z} \leq \boldsymbol{0},\ \nabla f(\boldsymbol{x}^*) = \boldsymbol{C}^\top \boldsymbol{z} = \sum_{k=1}^{|\mathcal{A}|} \boldsymbol{a}_{\mathcal{A}_k} z_k.$$

where $\mathcal{A}_k$ denotes k-th element in $\mathcal{A}(\boldsymbol{x}^*)$. Define $\boldsymbol{y} \in \mathbb{R}^m$: $y_i = -z_k$ for $i = \mathcal{A}_k$, and $y_i = 0$ for $i \in \mathcal{I}(\boldsymbol{x}^*)$. Then

$$\boldsymbol{y} \geq \boldsymbol{0},\ \nabla f(\boldsymbol{x}^*) = -\boldsymbol{A}^\top \boldsymbol{y},\ y_i(\boldsymbol{a}_i^\top \boldsymbol{x}^* - b_i) = 0,\ i = 1, \ldots, m.$$

Problems with Linear Equality Constraints

As a consequence, the first-order conditions for the problem

$$\text{minimize}_{\boldsymbol{x}} \; f(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \tag{2}$$

are given by:

Theorem: Linear Equality Constraints

If $\boldsymbol{x}^*$ is a local minimum of (2), then there is some $\boldsymbol{y} \in \mathbb{R}^m$ with

$$\nabla f(\boldsymbol{x}^*) = \boldsymbol{A}^\top \boldsymbol{y}.$$

Proof: This is a direct corollary of the previous theorem.

$$Ax = b \Leftrightarrow \begin{cases} Ax \le b \\ Ax \ge b \end{cases} \Leftrightarrow \begin{bmatrix} A \\ -A \end{bmatrix} x \le \begin{bmatrix} b \\ -b \end{bmatrix}$$

By previous theorem, there exists $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \ge 0$ s.t

$$\begin{cases} \nabla f(x^*) + \begin{bmatrix} A^T & -A^T \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 0 \Leftrightarrow \nabla f(x^*) + A^T \underbrace{(y_1 - y_2)}_{y \triangleq y_1 - y_2} = 0 \\ y_1^i (a_i^T x^* - b_i) = 0 \\ y_2^i (b_i - a_i^T x^*) = 0 \end{cases}$$

always holds as long as x^* is feasible

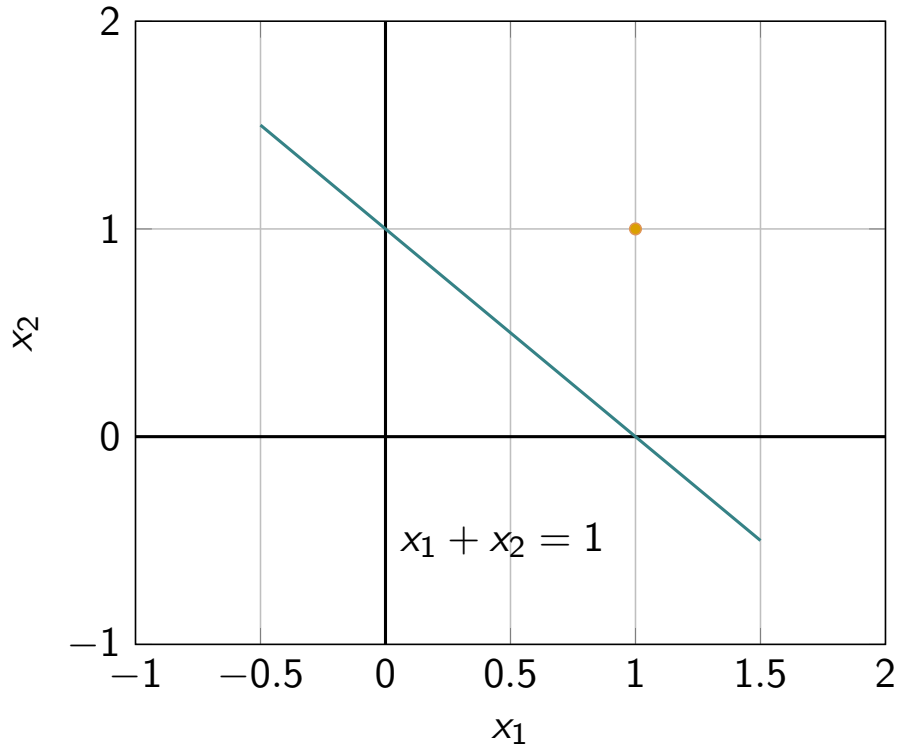
□

Example

Consider the problem:

$$\begin{aligned} \text{minimize} \quad & (x_1 - 1)^2 + (x_2 - 1)^2 \\ \text{s.t.} \quad & x_1 + x_2 = 1 \end{aligned}$$

- This problem finds the nearest point on the line $x_1 + x_2 = 1$ to the point $(1, 1)$



Example: Continued

By the FONC, if $\boldsymbol{x} = (x_1, x_2)$ is a local minimizer, then there exists y such that

$$\boldsymbol{A}^\top y = \nabla f(\boldsymbol{x}).$$

Here, we have $\boldsymbol{A} = (1, 1)$ and $\nabla f(\boldsymbol{x}) = (2x_1 - 2, 2x_2 - 2)^\top$.

Thus, there exists y with

$$2x_1 - 2 = y \quad \text{and} \quad 2x_2 - 2 = y.$$

Example: Continued

By the FONC, if $\boldsymbol{x} = (x_1, x_2)$ is a local minimizer, then there exists y such that

$$\boldsymbol{A}^\top y = \nabla f(\boldsymbol{x}).$$

Here, we have $\boldsymbol{A} = (1, 1)$ and $\nabla f(\boldsymbol{x}) = (2x_1 - 2, 2x_2 - 2)^\top$.

Thus, there exists y with

$$2x_1 - 2 = y \quad \text{and} \quad 2x_2 - 2 = y.$$

Combined with the constraint $x_1 + x_2 = 1$, we can infer:

$$x_1 = x_2 = \frac{1}{2}, \quad y = -1.$$

This is the only candidate for a local solution.

$\rightsquigarrow$ Indeed, it is a local minimizer (also a global minimizer).

Example: Linearly Constrained Least Squares

Consider the linearly constrained least squares problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{\beta}\in\mathbb{R}^n} \quad & ||\boldsymbol{X}\boldsymbol{\beta}-\boldsymbol{y}||_2^2 \\ \text{subject to} \quad & \boldsymbol{W}\boldsymbol{\beta}=\boldsymbol{\xi}, \end{aligned}$$

where $\boldsymbol{X}\in\mathbb{R}^{m\times n}$ and $\boldsymbol{W}\in\mathbb{R}^{p\times n}$.

- The gradient of the objective function is $2\boldsymbol{X}^\top(\boldsymbol{X}\boldsymbol{\beta}-\boldsymbol{y})$.
- By FONC, there exists dual variables $\boldsymbol{z}$ such that

$$\boldsymbol{W}^\top\boldsymbol{z}=2(\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta}-\boldsymbol{X}^\top\boldsymbol{y}).$$

Example: Linearly Constrained Least Squares

Consider the linearly constrained least squares problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{\beta}\in\mathbb{R}^n} \quad & ||\boldsymbol{X}\boldsymbol{\beta}-\boldsymbol{y}||_2^2 \\ \text{subject to} \quad & \boldsymbol{W}\boldsymbol{\beta}=\boldsymbol{\xi}, \end{aligned}$$

where $\boldsymbol{X}\in\mathbb{R}^{m\times n}$ and $\boldsymbol{W}\in\mathbb{R}^{p\times n}$.

- The gradient of the objective function is $2\boldsymbol{X}^\top(\boldsymbol{X}\boldsymbol{\beta}-\boldsymbol{y})$.
- By FONC, there exists dual variables $\boldsymbol{z}$ such that

$$\boldsymbol{W}^\top\boldsymbol{z}=2(\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta}-\boldsymbol{X}^\top\boldsymbol{y}).$$

- Together with feasibility, an optimal $\boldsymbol{\beta}$ must satisfy:

$$\exists \boldsymbol{z} \text{ such that } \boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta}=\frac{1}{2}\boldsymbol{W}^\top\boldsymbol{z}+\boldsymbol{X}^\top\boldsymbol{y}, \quad \boldsymbol{W}\boldsymbol{\beta}=\boldsymbol{\xi}.$$

Example Continued

We can write this FONC compactly:

$$\begin{bmatrix} \boldsymbol{W} & \boldsymbol{0}_{p\times p} \\ \boldsymbol{X}^\top \boldsymbol{X} & -\frac{1}{2}\boldsymbol{W}^\top \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta} \\ \boldsymbol{z} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\xi} \\ \boldsymbol{X}^\top \boldsymbol{y} \end{bmatrix}.$$

- The left-hand side matrix is of dimension $(n+p)\times(n+p)$.
- This is a system of linear equations with $n+p$ equations and $n+p$ unknowns. Solving this equation will yield the candidates for local minimizer
- If the left hand side matrix is full-rank, then there exists a unique solution.

Optimality Conditions: General Inequality Constraints

No Feasible Descent

We consider the nonlinear program:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \leq 0, \quad \forall\ i = 1, ..., m. \end{aligned} \tag{3}$$

- f and g_i, $i = 1, ..., m$, are assumed to be continuously differentiable

No Feasible Descent

We consider the nonlinear program:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \leq 0, \quad \forall\ i = 1, ..., m. \end{aligned} \tag{3}$$

- f and g_i, $i = 1, ..., m$, are assumed to be continuously differentiable

We interpreted the abstract FONC as: There is no feasible descent direction. Let us formalize this view below.

Lemma: No Feasible Descent for Inequality Constraints

Let $\boldsymbol{x}^*$ be a local minimum of (3). Then, there does not exist a vector $\boldsymbol{d} \in \mathbb{R}^n$ such that:

$$\underbrace{\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0}_{\text{descent direction}} \quad \text{and} \quad \underbrace{\nabla g_i(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \forall i \in \mathcal{A}(\boldsymbol{x}^*).}_{\text{feasible direction}}$$

Proof

Assume there is a $\boldsymbol{d}$ such that

$$\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \text{and} \quad \nabla g_i(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \forall i \in \mathcal{A}(\boldsymbol{x}^*).$$

By Taylor's theorem,

$$f(\boldsymbol{x} + t\boldsymbol{d}) = f(\boldsymbol{x}) + t\nabla f(\boldsymbol{x})^\top \boldsymbol{d} + o(t\|\boldsymbol{d}\|).$$

Since $\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0$, there exists $\bar{t} > 0$ such that

$$f(\boldsymbol{x}^* + t\boldsymbol{d}) < f(\boldsymbol{x}^*) \text{ for all } t \in (0, \bar{t}).$$

Consider $i \in \mathcal{A}(\boldsymbol{x}^*)$. By Taylor's theorem,

$$g_i(\boldsymbol{x} + t\boldsymbol{d}) = g_i(\boldsymbol{x}) + t\nabla g_i(\boldsymbol{x})^\top \boldsymbol{d} + o(t\|\boldsymbol{d}\|).$$

Since $\nabla g_i(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0$, there exists $\bar{\alpha}_i$ such that

$$g_i(\boldsymbol{x}^* + t\boldsymbol{d}) < g_i(\boldsymbol{x}^*) = 0 \text{ for all } t \in (0, \bar{\alpha}_i).$$

Proof Continued

Consider $i \in \mathcal{I}(\boldsymbol{x}^*)$. We have $g_i(\boldsymbol{x}^*) < 0$. By Taylor's theorem,

$$g_i(\boldsymbol{x} + t\boldsymbol{d}) = g_i(\boldsymbol{x}) + t\nabla g_i(\boldsymbol{x})^\top \boldsymbol{d} + o(t\|\boldsymbol{d}\|).$$

Since $g_i(\boldsymbol{x}^*) < 0$, we can let t small enough such that the sum of last two terms is smaller than $-g_i(\boldsymbol{x}^*)$. That means there exists β_i such that

$$g_i(\boldsymbol{x}^* + t\boldsymbol{d}) < g_i(\boldsymbol{x}^*) - g_i(\boldsymbol{x}^*) = 0 \text{ for all } t \in (0, \bar{\beta}_i).$$

If we pick $t \leq \min\left\{\bar{t}, \{\bar{\alpha}_i\}_{i\in\mathcal{A}(\boldsymbol{x}^*)}, \{\bar{\beta}_i\}_{i\in\mathcal{I}(\boldsymbol{x}^*)}\right\}$, then $f(\boldsymbol{x}^* + t\boldsymbol{d}) < f(\boldsymbol{x}^*)$ and $\boldsymbol{x}^* + t\boldsymbol{d}$ is feasible. This contradicts with the fact that $\boldsymbol{x}^*$ is a local minimum.

Remark: in this proof, we have shown any d s.t $\nabla g_i(x^*)^T d_i < 0$ for all $i \in \mathcal{A}(x^*)$ is a feasible direction at x^*.

Theorem: Fritz-John Conditions

Let $\boldsymbol{x}^*$ be a local minimum of (3). Then, there exists $\lambda_0 \geq 0$, $\lambda_1 \geq 0, \ldots, \lambda_m \geq 0$, which are not all zeros, such that

$$\lambda_0 \nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0,$$

$$\lambda_i \cdot g_i(\boldsymbol{x}^*) = 0, \quad \forall\ i = 1, \ldots, m.$$

$\lambda_0 \nabla f(x^*)$ can be represented as linear combination of $\{\nabla g_i(x^*)\}_i$

↑ complementary slackness

Theorem: Fritz-John Conditions

Let $\boldsymbol{x}^*$ be a local minimum of (3). Then, there exists $\lambda_0 \geq 0$, $\lambda_1 \geq 0$, ..., $\lambda_m \geq 0$, which are not all zeros, such that

$$\lambda_0 \nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0,$$

$$\lambda_i \cdot g_i(\boldsymbol{x}^*) = 0, \quad \forall\ i = 1, ..., m.$$

Major Drawback: The choice $\lambda_0 = 0$ is allowed.

- In this case, the Fritz-John conditions just impose linear dependence of the vectors $\{\nabla g_i(\boldsymbol{x}^*)\}_{i \in \mathcal{A}(\boldsymbol{x}^*)}$.

$\rightsquigarrow$ No information about f. Many points might satisfy this condition without being minimizer.

Proof

By the last Lemma,

$$\nexists \boldsymbol{d} : \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \text{and} \quad \nabla g_i(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \forall i \in \mathcal{A}(\boldsymbol{x}^*).$$

Proof

By the last Lemma,

$$\nexists \boldsymbol{d} : \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \text{and} \quad \nabla g_i(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \forall i \in \mathcal{A}(\boldsymbol{x}^*).$$

Consider the problem:

$$\min_{\boldsymbol{d},t} \; t \quad \text{s.t.} \quad \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \leq t, \; \nabla g_i(\boldsymbol{x}^*)^\top \boldsymbol{d} \leq t, \; \forall i \in \mathcal{A}(\boldsymbol{x}^*).$$

Then the optimal value needs to be 0. (why?)

Proof

By the last Lemma,

$$\nexists \boldsymbol{d} : \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \text{and} \quad \nabla g_i(\boldsymbol{x}^*)^\top \boldsymbol{d} < 0 \quad \forall i \in \mathcal{A}(\boldsymbol{x}^*). \quad \langle a \rangle$$

Consider the problem:

$$\min_{\boldsymbol{d},t} \; t \quad \text{s.t.} \quad \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \le t, \; \nabla g_i(\boldsymbol{x}^*)^\top \boldsymbol{d} \le t, \; \forall i \in \mathcal{A}(\boldsymbol{x}^*).$$

Then the optimal value needs to be 0. (why?) → (1) by ⟨a⟩, there is no feasible point if $t<0$

(2) $d=0$, $t=0$, then objective $=0$

Define

$$\boldsymbol{C} = \left(\boldsymbol{c}_i^\top\right)_{i \in \mathcal{A}(\boldsymbol{x}^*)}, \text{where } \boldsymbol{c}_i = \nabla g_i(\boldsymbol{x}^*) \text{ for } i \in \mathcal{A}(\boldsymbol{x}^*).$$

Then we have

$$0 = \min_{\boldsymbol{d},t} \; t \quad \text{s.t.} \quad t - \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \ge 0, \; t\mathbf{1} - \boldsymbol{C}\boldsymbol{d} \ge 0.$$

Proof — Continued

Taking the dual,

$$\max_{\lambda_0, \boldsymbol{\mu}} \ 0, \quad \text{s.t.} \quad \lambda_0, \boldsymbol{\mu} \geq 0, \ -\nabla f(\boldsymbol{x}^*)\lambda_0 - \boldsymbol{C}^\top \boldsymbol{\mu} = 0, \ \lambda_0 + \mathbf{1}^\top \boldsymbol{\mu} = 1,$$

Proof — Continued

Taking the dual,

$$\max_{\lambda_0, \boldsymbol{\mu}} \ 0, \quad \text{s.t.} \quad \lambda_0, \boldsymbol{\mu} \geq 0, \ -\nabla f(\boldsymbol{x}^*)\lambda_0 - \boldsymbol{C}^\top \boldsymbol{\mu} = 0, \ \lambda_0 + \mathbf{1}^\top \boldsymbol{\mu} = 1,$$

By strong duality, there must exists feasible $(\lambda_0, \boldsymbol{\mu})$ to the dual problem. Define $\boldsymbol{\lambda} \in \mathbb{R}^m$ such that $\lambda_i = \mu_k$ for $i = \mathcal{A}_k$, where $\mathcal{A}_k$ denotes k-th element in $\mathcal{A}(\boldsymbol{x}^*)$, and $\lambda_i = 0$ for $i \in \mathcal{I}(\boldsymbol{x}^*)$:

$$\lambda_0 \nabla f(\boldsymbol{x}^*) + \sum_{i=1}^m \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0,$$

$$\lambda_i g_i(\boldsymbol{x}^*) = 0, \quad i = 1, \ldots, m,$$

$$\lambda_0 + \sum_{i=1}^m \lambda_i = 1 \text{ (not all are zeros)},$$

$$\lambda_0, \boldsymbol{\lambda} \geq 0.$$

More General Cases: KKT Conditions

We have discussed Fritz-John Conditions (FONC) for general nonlinear inequality constraints.

- The critical flaw is that it can allow $\lambda_0 = 0$.
- We want to extend them to more useful cases, i.e, KKT conditions, which impose further assumptions to eliminate the possibility of $\lambda_0 = 0$.
- KKT conditions is the first-order necessary conditions (FONC) for general nonlinear optimization problems.
- The KKT conditions were originally named after H. Kuhn and A. Tucker, who first published the conditions in 1951. Later scholars discovered that the conditions had been stated by W. Karush in his master's thesis in 1939.

KKT Conditions: Inequality Constraints

KKT Conditions for Inequality Constraints

Recall the nonlinear optimization:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \leq 0, \quad \forall\ i = 1, ..., m. \end{aligned} \tag{4}$$

Theorem: KKT Conditions for Inequality Constrained Problems

Let $\boldsymbol{x}^*$ be a local minimum of (4) and suppose that the vectors

$$\{\nabla g_i(\boldsymbol{x}^*) : i \in \mathcal{A}(\boldsymbol{x}^*)\}$$

are linearly independent. Then, there exist $\lambda_1, ..., \lambda_m \geq 0$ such that

$$\begin{aligned} \nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0, \\ \lambda_i \cdot g_i(\boldsymbol{x}^*) = 0, \quad i = 1, ..., m. \end{aligned}$$

By Fritz-John Conditions, assume that $\lambda_0 = 0$, then

$$\sum_{i \in \mathcal{A}(\boldsymbol{x}^*)} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0.$$

Proof

By Fritz-John Conditions, assume that $\lambda_0 = 0$, then

$$\sum_{i \in \mathcal{A}(\boldsymbol{x}^*)} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0.$$

Since $\{\nabla g_i(\boldsymbol{x}^*) : i \in \mathcal{A}(\boldsymbol{x}^*)\}$ are linearly independent, then $\lambda_i = 0$ for $i \in \mathcal{A}(\boldsymbol{x}^*)$.

By Fritz-John Conditions, assume that $\lambda_0 = 0$, then

$$\sum_{i \in \mathcal{A}(\boldsymbol{x}^*)} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0.$$

Since $\{\nabla g_i(\boldsymbol{x}^*) : i \in \mathcal{A}(\boldsymbol{x}^*)\}$ are linearly independent, then $\lambda_i = 0$ for $i \in \mathcal{A}(\boldsymbol{x}^*)$. Also, $\lambda_i = 0$ for $i \in \mathcal{I}(\boldsymbol{x}^*)$.

By Fritz-John Conditions, assume that $\lambda_0 = 0$, then

$$\sum_{i \in \mathcal{A}(\boldsymbol{x}^*)} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0.$$

Since $\{\nabla g_i(\boldsymbol{x}^*) : i \in \mathcal{A}(\boldsymbol{x}^*)\}$ are linearly independent, then $\lambda_i = 0$ for $i \in \mathcal{A}(\boldsymbol{x}^*)$. Also, $\lambda_i = 0$ for $i \in \mathcal{I}(\boldsymbol{x}^*)$. Therefore, $\lambda_0, \lambda_1, \ldots, \lambda_m = 0$, contradicting the fact that $\lambda_0, \lambda_1, \ldots, \lambda_m$ are not all zeros in FJ conditions.

By Fritz-John Conditions, assume that $\lambda_0 = 0$, then

$$\sum_{i \in \mathcal{A}(\boldsymbol{x}^*)} \lambda_i \nabla g_i(\boldsymbol{x}^*) = 0.$$

Since $\{\nabla g_i(\boldsymbol{x}^*) : i \in \mathcal{A}(\boldsymbol{x}^*)\}$ are linearly independent, then $\lambda_i = 0$ for $i \in \mathcal{A}(\boldsymbol{x}^*)$. Also, $\lambda_i = 0$ for $i \in \mathcal{I}(\boldsymbol{x}^*)$. Therefore, $\lambda_0, \lambda_1, \ldots, \lambda_m = 0$, contradicting the fact that $\lambda_0, \lambda_1, \ldots, \lambda_m$ are not all zeros in FJ conditions.

Then, we conclude that $\lambda_0 > 0$, dividing two sides by λ_0 gives the results.

Today's lecture

- Fritz-John Condition
- KKT Condition for Inequality Constrained Problems

Next lecture

- More about KKT Condition

Questions?